

Frédéric Andrian

Software Engineer · Generative AI Platform Tooling

Bordeaux, France

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Professional Summary

Software Engineer completing an MSc-equivalent engineering degree in Artificial Intelligence and Cognitive Science. At INRIA (French National Institute for Research in Digital Science and Technology), I independently delivered an end-to-end Generative AI application from a user need to a functional solution, including Python and FastAPI services, LLM access, agent tools, RAG, model evaluation, a web interface, Dockerized components, and GitHub CI/CD. This experience developed my interest in the frameworks, services, and evaluation tools that help developers build reliable GenAI applications. Available from November 2026, I am motivated to work abroad to discover new working methods and grow within an international environment.

Experience

INRIA (French National Institute for Research in Digital Science and Technology), Mnémosyne Team

Bordeaux, France

AI Research Engineer Intern

Mar. 2026 – Sep. 2026

- Independently designed and developed, with guidance from my supervisors, an end-to-end multi-step GenAI application that helps researchers discover scientific papers.
- Built Python and FastAPI backend services, REST APIs documented with OpenAPI, and MongoDB persistence.
- Integrated the OpenAI API and designed the tools used by the agent to retrieve information and complete successive tasks.
- Built the agent without an orchestration framework to understand and control tool execution, state, memory, context, and multi-step behavior.
- Implemented the RAG lifecycle with document chunking, embeddings, semantic search, and a ChromaDB vector database.
- Managed a Dockerized web application for LLMs installed locally on a Mac Studio, including model installation and configuration.
- Benchmarked Qwen, Gemma, and GLM on an HPC cluster with Slurm across latency, TTFT, response quality, and context size.
- Evaluated multiple quantization configurations to identify the best trade-off between inference performance and response quality for each execution environment.
- Defined an experimental protocol, analyzed the results, and contributed to a scientific manuscript documenting the model evaluation.
- Dockerized selected components and implemented a GitHub CI/CD pipeline to automate integration.
- Evaluated OpenClaw in an isolated Docker environment with attention to the reliability and security of autonomous tool and file access.
- Developed a Python tool to automatically collect scientific paper metadata from multiple public sources.
- Implemented webhooks to relay the results of scheduled tasks.
- Developed a React and Next.js interface that enabled researchers to interact with the application and its tools.

Karunya University

Coimbatore, India

Reinforcement Learning Research Intern

Jun. 2025 – Sep. 2025

- Designed and compared deterministic path planning with a reinforcement learning approach for autonomous navigation of a hospital robot.
- Deployed and optimized the software and reinforcement learning model on an embedded Raspberry Pi.

Technical Skills

Programming and Backend GenAI Platform Tooling

Python, Java, FastAPI, Node.js, REST APIs, OpenAPI

OpenAI API, local LLM management, framework-free agents, tool design, memory and context management

RAG and Data LLM Evaluation

Chunking, embeddings, semantic search, ChromaDB, MongoDB, SQL

Qwen, Gemma, GLM, quantization, experimental protocols, latency, TTFT, response quality, context size

Operations and Deployment Frontend

Linux, Docker, GitHub CI/CD, Slurm, HPC environments, Raspberry Pi

React, Next.js

Machine Learning Development Tools Languages

PyTorch, Deep Learning, Reinforcement Learning, NLP

Git/GitHub, ChatGPT for AI-assisted development

French (native), English (professional, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP

Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science

2023 – 2026

- Coursework in software engineering, machine learning, large language models, natural language processing, deep learning, and human-computer interaction.